

Smart Building Platform Overview

Building Management System (BMS)
Integration Proposal — 15 March
2026

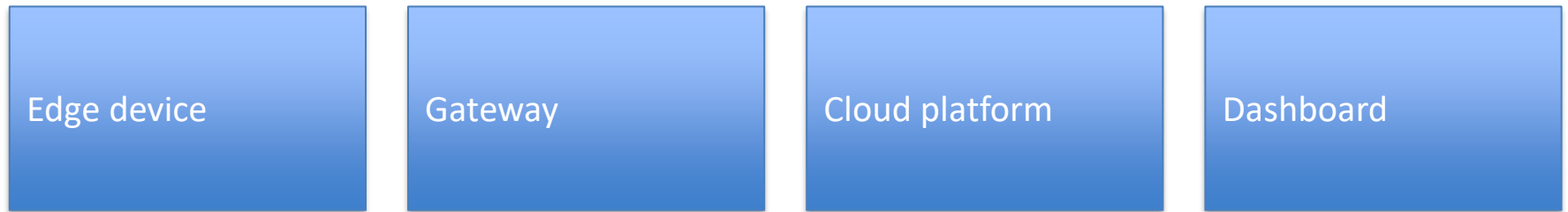
Building Management System

- The **Building Management System** monitors and controls HVAC equipment.
 - Air Handling Unit (AHU) and Fan Coil Unit (FCU) control
 - Variable Air Volume (VAV) terminal control
- **Alarm** and *equipment status* monitoring 24/7
- Energy Management System (EMS) for energy consumption analysis

Sensors and Field Devices

- Occupancy sensor
- CO2 sensor (0–2,000 ppm)
- Temperature sensor: ± 0.2 °C accuracy
- Humidity: 30–70 %RH
- Model **FX-PCG2611-0** controller, firmware v3.2.1

System Architecture



Field controllers communicate over BACnet/IP and Modbus TCP; the gateway publishes telemetry to the cloud platform via MQTT over TLS (port 8883).

Protocol Comparison

Protocol	Transport	Typical use	Default port
BACnet/IP	UDP	HVAC controllers	47808
Modbus TCP	TCP	Power meters, chillers	502
MQTT	TCP	IoT telemetry to cloud	1883 / 8883
KNX	Twisted pair / IP	Lighting and blinds	3671

Energy Efficiency and Demand Response

- Peak demand reduced by 18% in summer 2025
- Energy consumption: 1,250 MWh/year → 1,020 MWh/year
- **Demand response** signals are received from the utility via **OpenADR 2.0b**
- Setpoint adjustment: cooling setpoint raised from 24 °C to 26 °C during peak hours
- Power consumption is measured every 15 minutes

Network and Access

- BMS server: 192.168.1.100 (VLAN 20)
- Web dashboard:
<https://bms.example.com/dashboard>
- Support: support@example.com
- Configuration file: C:\Program Files\BMS\config\points.json
- REST API over HTTPS; payloads in JSON, legacy exports in XML and SQL

Operation and Maintenance

- When a chiller alarm occurs, the system automatically notifies the maintenance team by e-mail and creates a work order, and if the alarm is not acknowledged within 30 minutes, it is escalated to the facility manager.
- Scheduled maintenance of the AHU filters is performed every 3 months.
- The PLC and SCADA system remain in service during the migration period.

Building Automation Roadmap

- Phase 1 (Q2 2026): Building automation system upgrade
- Phase 2 (Q4 2026): IoT sensors and edge devices
- Phase 3 (2027): AI-based fault detection and diagnostics (FDD)